

AYE  
TECH HUB

• SOLAR ENERGY • LESSON 04

# Solar Panels — In Depth

I-V Curve · MPP · Temperature · Shading · Wiring

Lesson 04 of 30 | Solar Energy Program

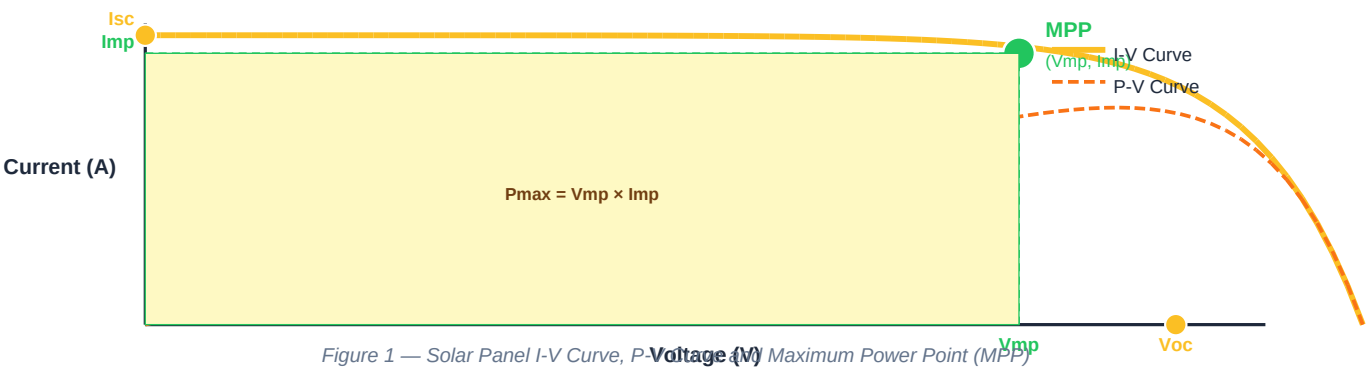
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- The I-V curve:  $I_{sc}$ ,  $V_{oc}$ ,  $V_{mp}$ ,  $I_{mp}$ , MPP and Fill Factor
- How MPPT tracks maximum power in real time
- Temperature derating — how much power you lose on a hot day
- Reading a panel datasheet — 7 fields with red-flag warnings
- Series vs parallel wiring, bypass diodes and shading losses

# The I-V Curve — A Panel's DNA

Every solar panel has a characteristic **Current-Voltage (I-V) curve** that describes how much current it delivers at every possible operating voltage. This curve is the most important piece of information on any panel datasheet. Everything else — energy yield, string sizing, inverter matching, temperature derating — follows from understanding it.



| Point / Region        | Symbol        | Definition                                                                                                 | Typical Value (400 W panel)                 |
|-----------------------|---------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| Short-Circuit Current | Isc           | Current when output terminals are shorted (V=0). Maximum current the panel can produce.                    | ~10.5 A                                     |
| Open-Circuit Voltage  | Voc           | Voltage when no current flows (open circuit). Maximum voltage across terminals.                            | ~41.5 V                                     |
| Maximum Power Current | Imp (or Impp) | Current at the Maximum Power Point — the operating point that yields peak power.                           | ~9.9 A                                      |
| Maximum Power Voltage | Vmp (or Vmpp) | Voltage at the Maximum Power Point.                                                                        | ~40.4 V                                     |
| Maximum / Peak Power  | Pmax (Wp)     | $P_{max} = V_{mp} \times I_{mp}$ . Rated at STC (1 000 W/m², 25°C, AM1.5).                                 | 400 Wp                                      |
| Fill Factor           | FF            | $FF = P_{max} / (V_{oc} \times I_{sc})$ . Measures how "square" the I-V curve is. Higher = better quality. | ~0.80 (good quality)                        |
| Knee Region           | —             | The rapid drop-off zone where voltage rises sharply but current falls — avoid operating here.              | Near Voc; bypass diodes protect this region |

**MPPT explained:** The inverter's MPPT tracker continuously adjusts the operating voltage of the string to stay exactly at Vmp — the peak of the P-V curve. As irradiance, temperature, and shading change throughout the day, Vmp shifts and the MPPT follows it in real time.

## Temperature Effects and the Panel Datasheet

Solar panels are rated at **STC — Standard Test Conditions**: 1 000 W/m<sup>2</sup> irradiance, 25°C cell temperature, AM1.5 spectrum. In the real world, panel temperatures regularly reach 50–70°C on a hot summer day. This causes a significant drop in output that every system designer must account for.

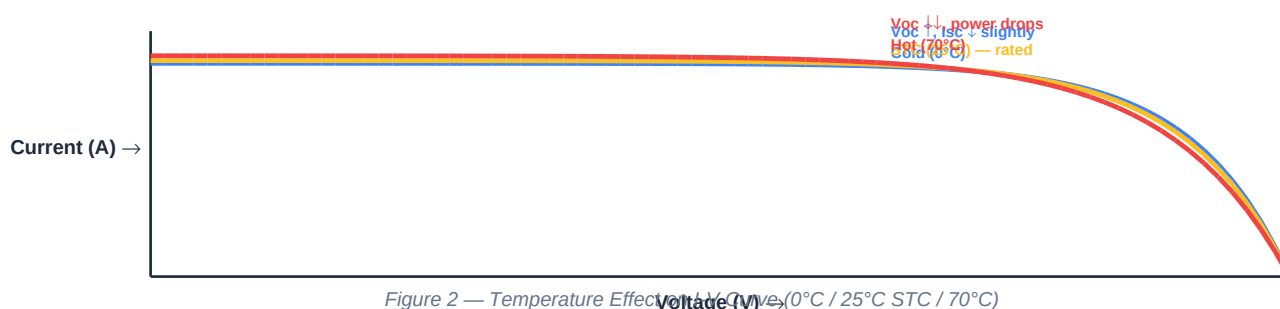


Figure 2 — Temperature Effect on I-V Curve (0°C / 25°C STC / 70°C)

| Temperature Coefficient            | Typical Value             | Effect at 65°C Cell Temp (vs 25°C STC)                                                  |
|------------------------------------|---------------------------|-----------------------------------------------------------------------------------------|
| Power (Pmax) — $\gamma$            | –0.35 to –0.45 %/°C       | 40°C above STC → –14% to –18% power loss A 400 W panel produces only 332–344 W          |
| Voltage (Voc) — $\beta$            | –0.27 to –0.36 %/°C       | Voc drops → affects string voltage for inverter MPPT range                              |
| Current (Isc) — $\alpha$           | +0.04 to +0.06 %/°C       | Isc rises slightly — small benefit; does not offset voltage drop                        |
| NOCT (Nominal Operating Cell Temp) | 42–48°C (varies by panel) | Cell temp = Ambient + (NOCT – 20°C) × (G / 800) Use for realistic daily yield estimates |

### How to Read a Solar Panel Datasheet

| Datasheet Field              | What to Check                                                      | Red Flags                                                                    |
|------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------|
| STC Power (Wp) and tolerance | Panel wattage; accept +3%/–0% tolerance (positive tolerance)       | Negative tolerance (–3%) means delivered power could be below rated          |
| Efficiency (%)               | Power per unit area. >21% = premium mono; 16–18% = entry-level     | Very cheap panels with claimed >23% efficiency — verify with test cert.      |
| Voc and Vmp                  | Use Voc × 1.15 safety factor for inverter max DC voltage check     | String Voc must never exceed inverter max input voltage (e.g. 1 000 V)       |
| Isc and Imp                  | Size fuses and DC cable to 1.25 × Isc per string                   | If combiner box fuses are undersized, short circuit can cause fire           |
| Temperature coefficients     | Lower (less negative) $\gamma$ = better in hot climates            | Cheap panels often omit NOCT — assume 48°C if not stated                     |
| Degradation / Warranty       | Linear power warranty: ≤0.5%/yr degradation to 80% at year 25      | Step-function warranty (e.g. 90% at yr 10, 80% at yr 25) is industry minimum |
| IEC/UL Certifications        | IEC 61215 (performance) + IEC 61730 (safety) are mandatory minimum | No third-party certification = significant quality and warranty risk         |

**Bypass diodes: Each panel contains 3 bypass diodes (one per cell group). When a cell is shaded, its diode activates and routes current around the shaded group — preventing the "Christmas lights effect" where one shaded cell kills the whole string. The shaded section loses ~33% of panel power, not 100%.**

## Wiring Panels: Series, Parallel and Shading

How you connect panels determines the voltage and current delivered to the inverter. Getting this right is critical for inverter compatibility, energy yield, and safety. Two rules always hold: **series adds voltage**, **parallel adds current**. Power is always  $V \times I$ .

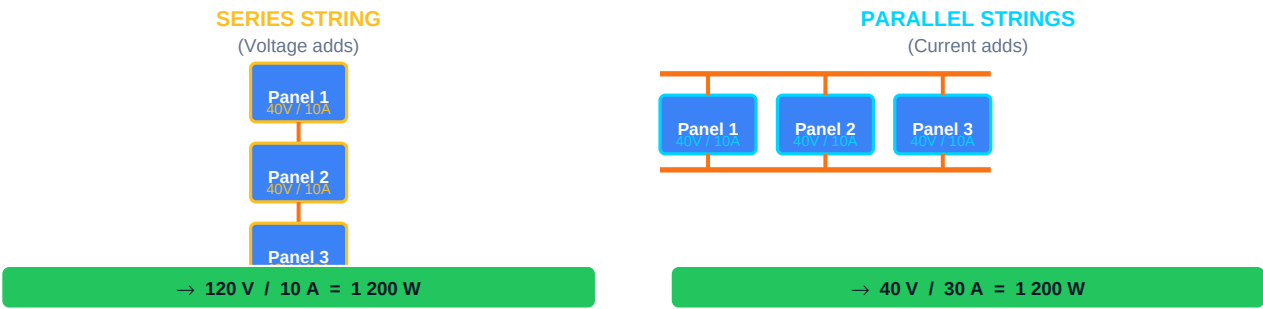


Figure 3 — Series Wiring Adds Voltage; Parallel Wiring Adds Current

| Wiring                                        | Voltage                      | Current                     | Power   | When to Use                                                               |
|-----------------------------------------------|------------------------------|-----------------------------|---------|---------------------------------------------------------------------------|
| 3 panels in series (40V / 10A each)           | $3 \times 40 = 120\text{ V}$ | 10 A                        | 1 200 W | Higher voltage strings reduce cable losses; match inverter MPPT range     |
| 3 panels in parallel (40V / 10A each)         | 40 V                         | $3 \times 10 = 30\text{ A}$ | 1 200 W | Low-voltage battery systems; when strings must be short                   |
| 2S $\times$ 2P (2 series $\times$ 2 parallel) | $2 \times 40 = 80\text{ V}$  | $2 \times 10 = 20\text{ A}$ | 1 600 W | Balance voltage and current; used in combiner boxes with multiple strings |

### Shading Losses — The Biggest Silent Killer

| Shading Scenario                                   | Loss Without Bypass Diodes             | Loss With Bypass Diodes                           | Solution                                                         |
|----------------------------------------------------|----------------------------------------|---------------------------------------------------|------------------------------------------------------------------|
| 1 cell fully shaded in a 60-cell panel             | Up to 100% of string power lost        | ~33% of panel power (1 of 3 cell groups bypassed) | Bypass diodes (standard on all modern panels)                    |
| 1 panel shaded in a 10-panel string (no optimizer) | 50–80% string power loss               | 10–30% string power loss (bypass diodes active)   | Microinverters or DC power optimisers per panel                  |
| Partial shade from nearby building (AM/PM)         | Year-round loss of 15–25% energy yield | Reduced with optimisers                           | Reorient array; use east-west split; shade analysis in PVsyst    |
| Bird droppings / soiling on 1 panel                | Same as 1 panel shaded                 | Same as 1 panel shaded                            | Regular cleaning; tilt $\geq 10^\circ$ for self-cleaning by rain |

### Panel Selection Decision Guide

| Priority                           | Best Technology                                                             | Why                                                                          |
|------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Max output, limited roof space     | Monocrystalline 420–440 Wp (half-cut, multi-busbar)                         | Highest efficiency (21–23%); best W/m <sup>2</sup> ; lowest BOS cost per kWp |
| Hot climate performance            | Mono with low temperature coefficient ( $\gamma < -0.35\%/^\circ\text{C}$ ) | Less power lost on hot days; bifacial adds ground-reflected gain             |
| Shade-affected roof                | Microinverters or DC optimisers + any good mono panel                       | Panel-level MPPT eliminates string mismatch; worth the extra cost            |
| Large ground-mount (cost priority) | Bifacial mono, large format (480–580 Wp), tracker optional                  | Bifacial adds 5–15% yield; large format lowers installation labour           |

|                    |                                            |                                                                      |
|--------------------|--------------------------------------------|----------------------------------------------------------------------|
| Budget residential | Standard 400–410 Wp mono + string inverter | Best value; reliable; 10+ year proven track record; widely available |
|--------------------|--------------------------------------------|----------------------------------------------------------------------|

*What comes next — SOLAR-005: Solar Batteries — lead-acid vs LFP, state of charge, depth of discharge, battery sizing calculations, and how to specify the right battery bank for your system.*